Uganda Institute of Information and Communications Technology

"In Quest of Excellence"

BSM121: RESEARCH METHODS

MODULE 2: Types of Research

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Classification of Research

RESEARCH TYPES

1. Classification by Purpose
- Basic (Pure Fundamental)
- Applied
- Evaluation

2. Classification by Objective
- Exploratory
- Descriptive
- Explanatory/Causal
- Predictive
- Prescriptive/Design

3. Classification by Time Horizon
- Cross-sectional
- Longitudinal

4. Classification by Degree of Control
- Experimental
- Quasi-experimental
- Non-experimental

5. Classification by Data Source
- Survey
- Case Study
- Document Analysis
- Systematic Review

Classification by Purpose

a) Basic Research

- Description:
  - Focuses on expanding knowledge and understanding fundamental principles without immediate practical applications but only to the scholarly interest of a community of researchers.
  - Basic researchers are more detached and academic in their approach and tend to have their own motives.

- Purpose: Discovering general principles, theories, and laws governing natural and social phenomena.

- Examples:
  - A scientist studying the properties of subatomic particles to understand the building blocks of matter.
  - Studying how different learning theories to explain motivation in general

Classification by Purpose

b) Applied Research

- Description:
  - Also known as action research, operations research, social research, and decision-linked research
  - Seeks to find practical solutions for an immediate problem of society using existing knowledge.
- Purpose: Offers real-world solutions.
- Examples:
  - Neighborhood Vehicle Sharing AI Application
  - Developing a vaccine for a specific disease e.g. Covidex

Classification by Purpose

c) Evaluation Research

- **Purpose**: Assesses whether government programs or policies are achieving their intended goals, how efficiently they are implemented, and what impact they produce
- **Examples**:
  - Evaluation of the Impact of Uganda's National ICT Policy on Digital Transformation in Higher Education Institutions.
  - Evaluation of the Effectiveness of Universal Primary Education in Improving Literacy Levels in Uganda.
  - Evaluation of the Effectiveness of the Parish Development Model in Improving Household Incomes in Rural Uganda.

Classification by Objective

a) Exploration

- Focuses on investigating new or poorly understood topics or phenomena.
- It is particularly useful when researchers lack a clear idea (vague) of the problems they will meet during the course of the study.
- **Example:** Investigating the potential uses of Augmented and Virtual Reality (AVR) in education.

Classification by Objective

b) Description

- This describes characteristics, phenomena, processes, or relationships.
- Focus: "what is" rather than "why" or "how"
- Examples:
  - Documenting the student registration at UICT.
  - The Prevalence of Diabetes in Urban Versus Rural Communities
  - Usage Patterns of Mobile Applications Among Teenagers

Classification by Objective

c) Causal Explanation

- This goes beyond description and attempts to establish causes or reasons behind the observed phenomena.
- Example:
  - If a researcher finds that communities having higher family sizes have higher child death, s/he is performing a **descriptive study.**
  - If a researcher is explaining why it is so, s/he is performing an **explanatory study.**

Classification by Objective

d) Prediction

- Prediction seeks to forecast future occurrences based on current data or trends.

- **Example:** Predicting Uganda's population growth over the next 30 years to plan for housing and public services.

e) Prescriptive research

- Goes beyond describing or explaining a problem and proposes **specific solutions or actions to address it.**

- It answers the question: "What should be done to solve the problem?"

Classification Time Horizon

Type	Description	Example
Cross-Sectional Research	Data is collected at one point in time.	Survey of student satisfaction conducted in 2025.
Longitudinal Research	Data is collected over a long period to observe changes.	Tracking student performance from year 1 to graduation.

Classification Degree of Control

a) Experimental Research

- The researcher manipulates one or more independent variables to observe its effects on dependent variable.
- Presence of control and experiment groups
- Random assignment of subjects
- High level of control over the research environment
- Ability to establish cause-and-effect relationships
- Example
  - Suppose a researcher wants to test whether an AI-based learning system improves student performance.
  - Two groups of students are created:
    - Experimental group → uses the AI learning system
    - Control group → uses traditional teaching methods
    - If the experimental group performs better, the improvement can be attributed to the AI system.

Classification Degree of Control

b) Quasi-Experimental Research

- Resembles experimental research but lacks random assignment of subjects to groups.
- The researcher still manipulates variables but has moderate control over the research conditions.
- **Example**
  - Suppose a university introduces a new digital learning platform in one department but not another.
  - The researcher compares:
  - Department using the platform
  - Department not using the platform
  - Because students were **not randomly assigned**, it is considered quasi-experimental.

Classification Degree of Control

c) Non-experimental research

- The researcher does not manipulate variables but simply observes and analyzes existing conditions.
- This type has little or no control over variables.
- Example
  - A researcher studies the relationship between internet usage and academic performance among university students.
  - The researcher simply collects data using surveys and analyzes the relationship without manipulating any variables.

Classification by Data Sources

a) Survey Research

- Method of collecting data from a large group of people using questionnaires or structured interviews to gather information about opinions, behaviors, attitudes, or characteristics.

- Key Characteristics
  - Uses structured questionnaires
  - Data collected from many respondents
  - Often uses sampling techniques
  - Data is usually quantitative

- Example: A researcher studying student satisfaction with e-learning in universities distributes questionnaires to students across several institutions.

Classification by Data Sources

b) Case Study Research

- This is an in-depth investigation of a single individual, group, organization, institution, or event over a specific period of time.

- Key Characteristics
  - Focuses on one or a few cases
  - Uses multiple sources of data
  - Provides detailed and rich information
  - Often used in qualitative research

- Example: A researcher may conduct a case study of a single university implementing AI in teaching to understand how it affects learning outcomes.

Classification by Data Sources

c) Document Analysis Research

- This involves systematically reviewing and evaluating existing documents to extract meaningful information.

- Types of Documents Used
  - Government policies
  - Institutional reports
  - Research articles
  - Meeting minutes
  - Newspapers
  - Historical records

- Example: A researcher examining the impact of digital transformation in higher education may analyze policy documents, university strategic plans, and government ICT reports.

Classification by Data Sources

d) Systematic Review Research

- Involves collecting, evaluating, and synthesizing existing studies on a specific research question using a structured and transparent process.

- Key Characteristics
  - Follows a structured and rigorous methodology
  - Reviews many research studies
  - Uses clear inclusion and exclusion criteria
  - Often includes meta-analysis

- Example: A systematic review could examine studies on the use of machine learning in predicting student dropout in higher education.